

Supabase Storage dan Firebase Firestore

A. Membuat Account Supabase

1. Daftar / Sign-up ke Supabase apabila belum memiliki account Supabase
2. Buat organisasi baru di Supabase seperti gambar berikut ini dan kemudian klik **Create Organization**

Create a new organization

This is your organization within Supabase.
For example, you can use the name of your company or department.

Name
What's the name of your company or team?

Type
What would best describe your organization?

Plan
Pricing [↗](#)
The Plan applies to your new organization.

You can rename your organization later

3. Buat project baru seperti gambar berikut ini

Create a new project

Your project will have its own dedicated instance and full Postgres database.
An API will be set up so you can easily interact with your new database.

Organization
Free

Project name

Database password
This password is strong. [Generate a password.](#)

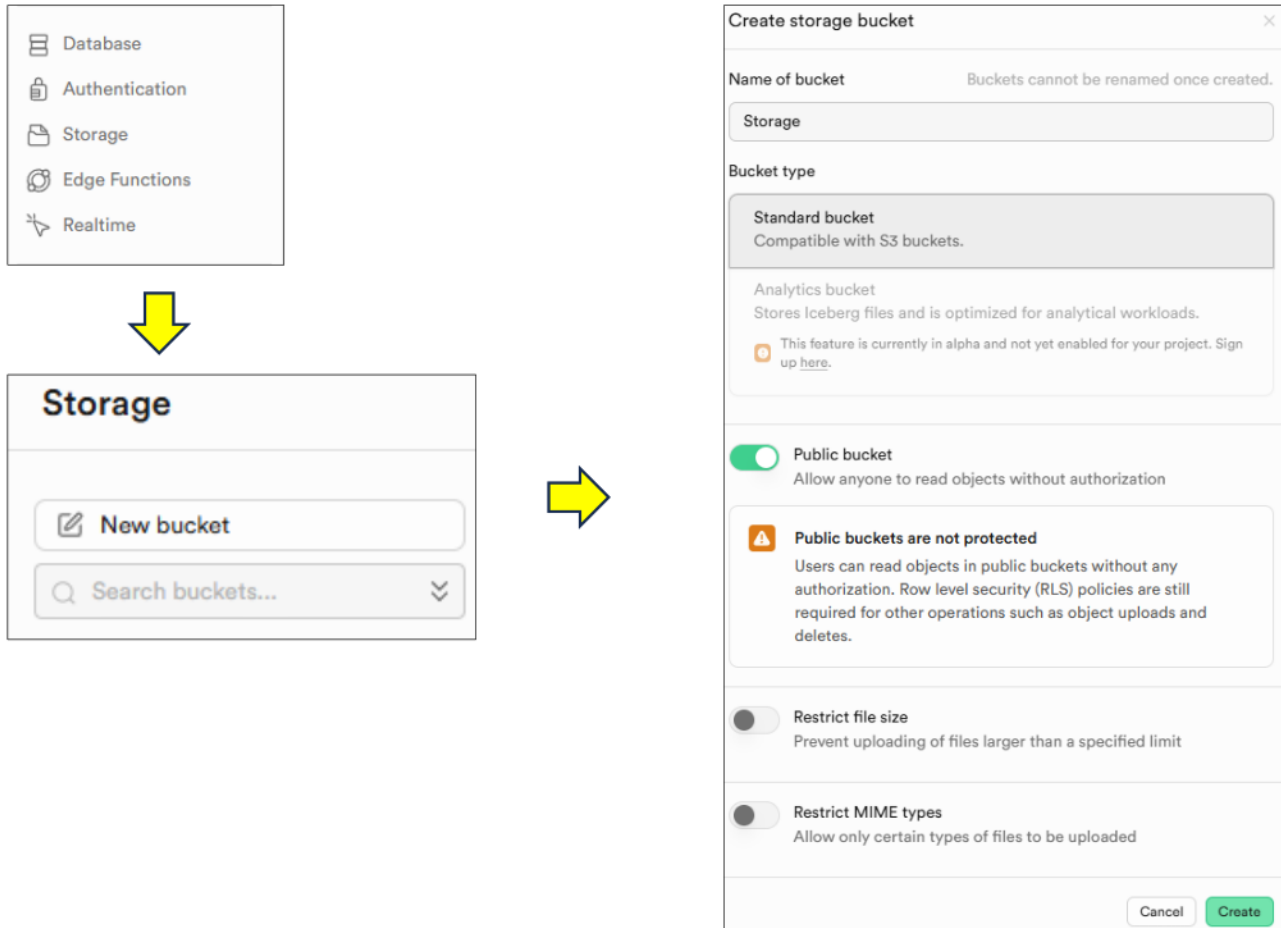
Region
Select the region closest to your users for the best performance.

[SECURITY OPTIONS >](#)

[ADVANCED CONFIGURATION >](#)

B. Membuat *Bucket* Sebagai *File Storage* di Supabase

4. Pada menu sebelah kiri halaman dashboard, klik pada menu **Storage** dan kemudian klik pada tombol **New Bucket**. Isikan **nama bucket** dan pastikan centang pada pilihan **Public Bucket** dan klik **create** seperti pada gambar dibawah ini



C. Menambahkan *Policies* di Bucket

Untuk dapat membaca dan menulis (*read* dan *write*) file pada bucket yang anda buat di Supabase maka anda memerlukan *policies* untuk membaca file dan menulis file. Untuk menambahkan *policies* tersebut ikuti langkah-langkah berikut ini :

5. Pilih/klik pada nama *bucket* yang tadi sudah anda buat
6. Klik pada menu **Policies**, kemudian klik pada tombol **New Policy** pada *bucket* anda, dan kemudian klik pada menu **For full customization**



7. Untuk menambahkan **akses membaca file (*read*)**, pada pilihan yang tersedia, isikan data seperti gambar berikut ini dan kemudian klik pada tombol **Review** dan **Save Policy**

Adding new policy to storage

Policy name

read file

A descriptive name for your policy

Allowed operation

☒ SELECT ☐ INSERT ☐ UPDATE ☐ DELETE

upload

download

list

update

move

copy

remove

createSignedUrl

createSignedUrls

getPublicUrl

Based on the operations you have selected, you can use the highlighted functions in the [client library](#).

Target roles

Defaults to all (public) roles if none selected

Apply policy to the selected roles

Policy definition

1 bucket_id = 'storage'

Provide a SQL conditional expression that returns a boolean.

8. Untuk menambahkan **akses menulis file (write)**, pada pilihan yang tersedia, isikan data seperti gambar berikut ini dan kemudian klik pada tombol **Review** dan **Save Policy**

Adding new policy to storage

Policy name

write file

A descriptive name for your policy

Allowed operation

☐ SELECT ☒ INSERT ☐ UPDATE ☐ DELETE

upload

download

list

update

move

copy

remove

createSignedUrl

createSignedUrls

getPublicUrl

Based on the operations you have selected, you can use the highlighted functions in the [client library](#).

Target roles

Defaults to all (public) roles if none selected

Apply policy to the selected roles

Policy definition

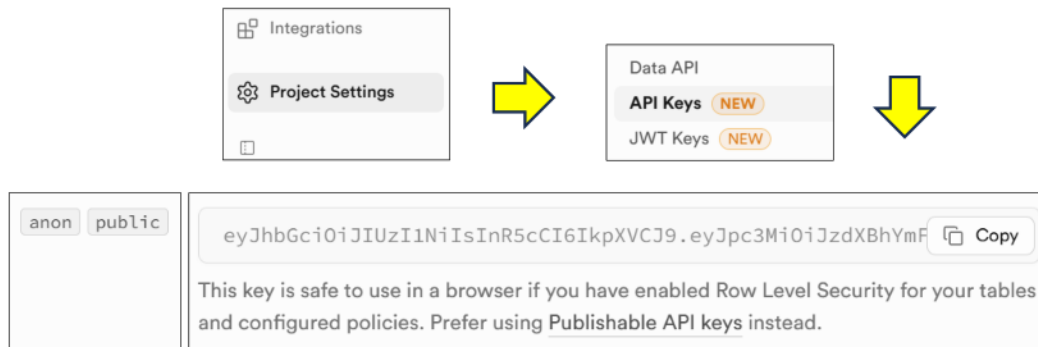
1 bucket_id = 'storage'

Provide a SQL conditional expression that returns a boolean.

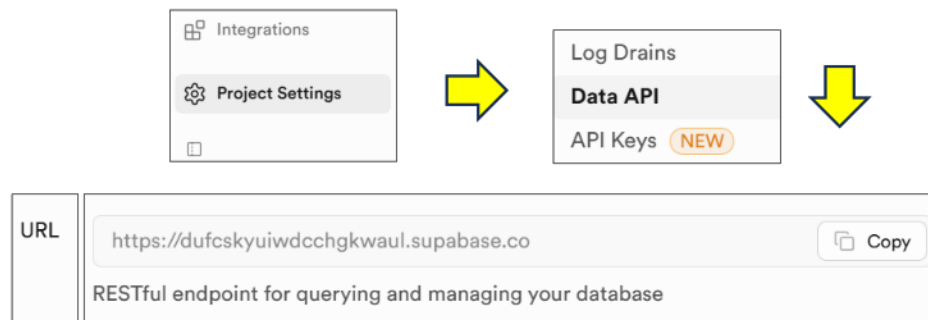
D. Mendapatkan API Key dan alamat URL client Supabase

Untuk mendapatkan API key dan alamat URL Supabase yang dibutuhkan oleh client Supabase, maka ikutilah langkah-langkah berikut ini

9. **API Key** bisa anda dapatkan melalui menu **Project Settings** → **API Keys** → salin API Keys **anon public** seperti pada gambar berikut ini



10. Alamat URL client Supabase bisa anda dapatkan melalui menu **Project Settings** → **Data API** → salin **URL** seperti pada gambar berikut ini



E. Mendaftarkan Package Aplikasi Android ke Firebase

11. Sebelum memulai membuat aplikasi Android, daftarkan dulu nama package aplikasi Android anda ke Firebase dan download file google-services.json untuk digunakan di project aplikasi Android anda

F. Membuat Aplikasi Android

12. Buat project baru aplikasi Android dengan nama package aplikasi sesuai dengan yang anda daftarkan ke Firebase
13. Salin file **google-services.json** yang anda dapatkan dari Firebase ke dalam project aplikasi anda
14. Pada file **AndroidManifest.xml**, tuliskan kode berikut ini

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">
    <uses-permission android:name="android.permission.INTERNET" />
    <uses-permission android:name="android.permission.ACCESS_NETWORK_STATE" />
    <uses-permission android:name="android.permission.ACCESS_WIFI_STATE" />

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportRtl="true"
        android:theme="@style/Theme.Material3.Light"/>
```

```
tools:targetApi="31">
<activity
    android:name=".MainActivity"
    android:exported="true">
    <intent-filter>
        <action android:name="android.intent.action.MAIN" />

        <category android:name="android.intent.category.LAUNCHER" />
    </intent-filter>
</activity>
</application>

</manifest>
```

15. Pada file **build.gradle.kts(Project)**, tuliskan kode berikut ini

build.gradle.kts (Project)

```
// Top-level build file where you can add configuration options common to all sub-
projects/modules.
plugins {
    alias(libs.plugins.android.application) apply false
    alias(libs.plugins.kotlin.android) apply false
    id("com.google.gms.google-services") version "4.4.3" apply false
}
```

16. Pada file **build.gradle.kts(Module)**, tuliskan kode berikut ini

build.gradle.kts (Module)

```
plugins {
    alias(libs.plugins.android.application)
    alias(libs.plugins.kotlin.android)
    id("com.google.gms.google-services")
}

android {
    namespace = "mi.polinema.appsupab"
    compileSdk = 36

    defaultConfig {
        applicationId = "mi.polinema.appsupab"
        minSdk = 26
        targetSdk = 36
        versionCode = 1
        versionName = "1.0"

        testInstrumentationRunner = "androidx.test.runner.AndroidJUnitRunner"
    }

    buildTypes {
        release {
            isMinifyEnabled = false
            proguardFiles(
                getDefaultProguardFile("proguard-android-optimize.txt"),
                "proguard-rules.pro"
            )
        }
    }
}
```

```
    }  
  }  
  
  buildFeatures {  
    viewBinding = true  
  }  
  
  compileOptions {  
    sourceCompatibility = JavaVersion.VERSION_23  
    targetCompatibility = JavaVersion.VERSION_23  
  }  
  kotlin {  
    jvmToolchain(23)  
  }  
}  
  
dependencies {  
  implementation("com.google.firebase:firebase-firestore:26.0.1")  
  implementation(platform("io.github.jan-tennert.supabase:bom:3.2.4"))  
  implementation("io.github.jan-tennert.supabase:storage-kt")  
  implementation("io.ktor:ktor-client-android:3.3.0")  
  implementation("com.squareup.picasso:picasso:2.8")  
  implementation(libs.androidx.core.ktx)  
  implementation(libs.androidx.appcompat)  
  implementation(libs.material)  
  implementation(libs.androidx.activity)  
  implementation(libs.androidx.constraintlayout)  
  testImplementation(libs.junit)  
  androidTestImplementation(libs.androidx.junit)  
  androidTestImplementation(libs.androidx.espresso.core)  
}
```

17. Pada file **libs.version.toml**, tuliskan kode berikut ini (optional)

libs.version.toml

```
[versions]  
agp = "8.13.0"  
kotlin = "2.2.20"  
coreKtx = "1.17.0"  
junit = "4.13.2"  
junitVersion = "1.3.0"  
espressoCore = "3.7.0"  
appcompat = "1.7.1"  
material = "1.13.0"  
activity = "1.11.0"  
constraintlayout = "2.2.1"  
  
[libraries]  
androidx-core-ktx = { group = "androidx.core", name = "core-ktx", version.ref = "coreKtx" }  
junit = { group = "junit", name = "junit", version.ref = "junit" }  
androidx-junit = { group = "androidx.test.ext", name = "junit", version.ref = "junitVersion" }  
androidx-espresso-core = { group = "androidx.test.espresso", name = "espresso-core",  
version.ref = "espressoCore" }  
androidx-appcompat = { group = "androidx.appcompat", name = "appcompat", version.ref =  
"appcompat" }  
material = { group = "com.google.android.material", name = "material", version.ref =  
"material" }  
androidx-activity = { group = "androidx.activity", name = "activity", version.ref = "activity" }
```

```
androidx.constraintlayout = { group = "androidx.constraintlayout", name =  
"constraintlayout", version.ref = "constraintlayout" }
```

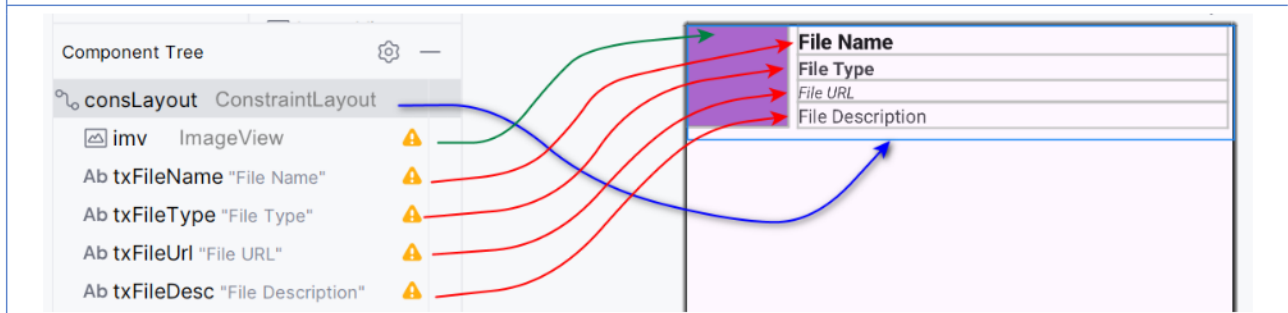
[plugins]

```
android-application = { id = "com.android.application", version.ref = "agp" }
```

```
kotlin-android = { id = "org.jetbrains.kotlin.android", version.ref = "kotlin" }
```

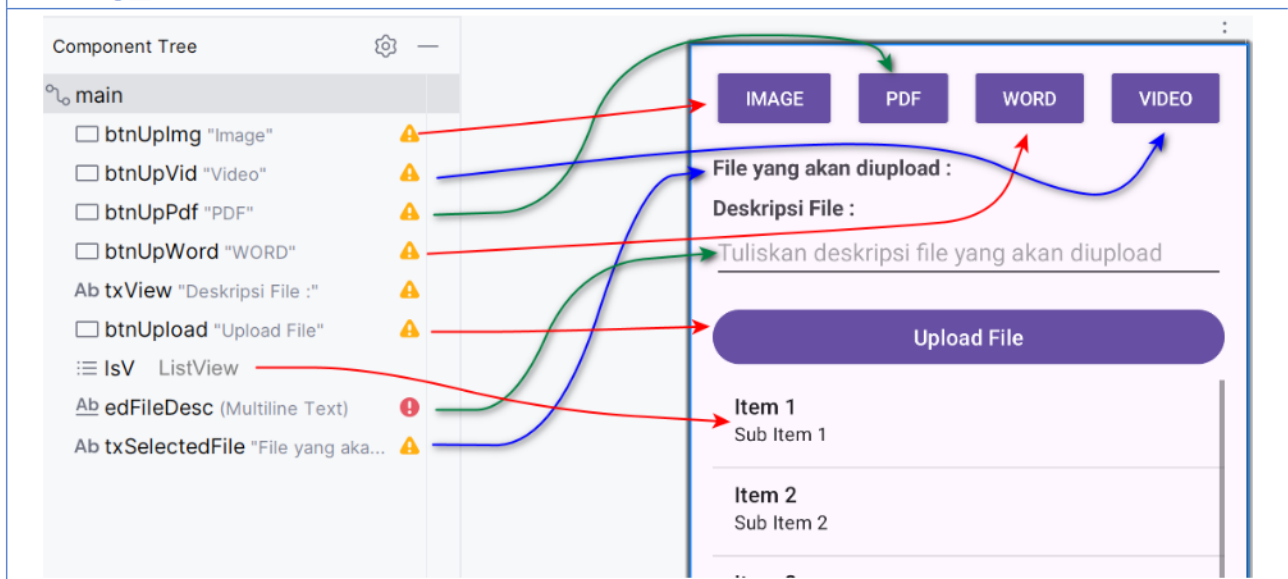
18. Buat layout dengan nama **row_data.xml** seperti pada gambar berikut ini

row_data.xml



19. Buat layout dengan nama **activity_main.xml** seperti pada gambar berikut ini

activity_main.xml



20. Buat sebuah Kotlin Class dengan nama **CustomAdapter.kt**

CustomAdapter.kt

```
package mi.polinema.appsupab  
import android.content.Context  
import android.content.Intent  
import android.net.Uri  
import android.view.LayoutInflater  
import android.view.View  
import android.view.ViewGroup  
import android.widget.BaseAdapter
```

```
import android.widget.ImageView
import android.widget.TextView
import androidx.constraintlayout.widget.ConstraintLayout
import androidx.core.content.ContextCompat
import com.squareup.picasso.Picasso
import androidx.core.net.toUri

class CustomAdapter(val context : Context,
                    arrayList : ArrayList<HashMap<String, Any>>) : BaseAdapter() {
    val F_NAME = "file_name"
    val F_TYPE = "file_type"
    val F_URL = "file_url"
    val F_DESC = "file_description"
    val list = arrayList
    var uri = Uri.EMPTY

    inner class ViewHolder(){
        var txFileName : TextView? = null
        var txFileType : TextView? = null
        var txFileUrl : TextView? = null
        var txFileDesc : TextView? = null
        var consLayout : ConstraintLayout? = null
        var imv : ImageView? = null
    }

    override fun getCount(): Int {
        return list.size
    }

    override fun getItem(position: Int): Any? {
        return list.get(position)
    }

    override fun getItemId(position: Int): Long {
        return 0
    }

    override fun getView(
        position: Int,
        convertView: View?,
        parent: ViewGroup?
    ): View? {
        var holder = ViewHolder()
        var view = convertView
        if(convertView == null){
            val inflater = context.getSystemService(Context.LAYOUT_INFLATER_SERVICE)
                as LayoutInflater

            view = inflater.inflate(R.layout.row_data,null,true)

            holder.txFileName = view!!.findViewById(R.id.txFileName) as TextView
            holder.txFileType = view!!.findViewById(R.id.txFileType) as TextView
            holder.txFileUrl = view!!.findViewById(R.id.txFileUrl) as TextView
            holder.txFileDesc = view!!.findViewById(R.id.txFileDesc) as TextView
            holder.consLayout = view!!.findViewById(R.id.consLayout) as ConstraintLayout
            holder.imv = view!!.findViewById(R.id.imv) as ImageView

            view.tag = holder
        }
    }
}
```

```
}else{
    holder = view!!.tag as ViewHolder
}

val fileType = list.get(position).get(F_TYPE).toString()
val fileDesc =
    if (list.get(position).get(F_DESC).toString().equals("null"))
        "No Description" else list.get(position).get(F_DESC).toString()
uri = list.get(position).get(F_URL).toString().toUri()

holder.txFileName!!.setText(list.get(position).get(F_NAME).toString())
holder.txFileType!!.setText(fileType)
holder.txFileUrl!!.setText(uri.toString())
holder.txFileDesc!!.setText(fileDesc)
holder.consLayout!!.setOnClickListener {
    val intent = Intent(Intent.ACTION_VIEW).setData(
        holder.txFileUrl!!.text.toString().toUri() )
    val originalColor = android.graphics.Color.TRANSPARENT
    it.setBackgroundColor(
        ContextCompat.getColor(context, android.R.color.holo_blue_light))
    it.postDelayed({it.setBackgroundColor(originalColor)}, 300)
    context.startActivity(intent)
}

when(fileType){
    "PDF" -> {holder.imv!!.setImageResource(android.R.drawable.ic_dialog_dialer)}
    "Docx" -> {holder.imv!!.setImageResource(android.R.drawable.ic_menu_edit)}
    "Video" -> {holder.imv!!.setImageResource(android.R.drawable.ic_media_play)}
    "Image" -> {
        Picasso.get().load(uri).into(holder.imv)}
}
return view!!
}
```

21. Pada file **MainActivity.kt**, tuliskan kode program seperti berikut ini

MainActivity.kt

```
package mi.polinema.appsupab

import android.annotation.SuppressLint
import android.content.Intent
import android.net.Uri
import android.os.Bundle
import android.provider.OpenableColumns
import android.util.Log
import android.view.View
import android.widget.Toast
import androidx.activity.enableEdgeToEdge
import androidx.appcompat.app.AlertDialog
import androidx.appcompat.app.AppCompatActivity
import androidx.core.view.ViewCompat
import androidx.core.view.WindowInsetsCompat
import androidx.lifecycle.lifecylceScope
import com.google.firebase.firestore.CollectionReference
import com.google.firebase.firestore.FirebaseFirestore
```

```
import io.github.jan.supabase.createSupabaseClient
import io.github.jan.supabase.exceptions.UnknownRestException
import io.github.jan.supabase.storage.Storage
import io.github.jan.supabase.storage.UploadStatus
import io.github.jan.supabase.storage.storage
import io.github.jan.supabase.storage.uploadAsFlow
import kotlinx.coroutines.launch
import mi.polinema.appsupab.databinding.ActivityMainBinding
import java.io.FileNotFoundException

class MainActivity : AppCompatActivity(), View.OnClickListener {
    //inisialisasi client Supabase
    val supabase = createSupabaseClient(
        supabaseUrl = "https://dufcskyuiwdcchgkwaul.supabase.co",
        supabaseKey = "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpc" +
            "3MiOiJzdXBhYmFzZSI6ImFub24iLCJpYXQiOiE3NTkxNTUzM" +
            "t2t3YXVsIiwicm9sZSI6ImFub24iLCJpYXQiOiE3NTkxNTUzM" +
            "zMsImV4cCI6IjE6MjA3NDczMTMzM30iUNZkTHE6axUn9A7S" +
            "yGEwo" +
            "VY1CSKfbjgk5wLxIamXNM"
    ) {
        install(Storage) //menggunakan Supabase Storage
    }

    lateinit var b: ActivityMainBinding
    lateinit var alFile: ArrayList<HashMap<String, Any>>
    lateinit var adapter: CustomAdapter
    lateinit var uri: Uri
    lateinit var db : CollectionReference
    val F_NAME = "file_name"
    val F_TYPE = "file_type"
    val F_URL = "file_url"
    val F_DESC = "file_description"
    val RC_OK = 100
    var fileType = ""
    var fileName = ""
    lateinit var dialog : AlertDialog

    //menampilkan data dari Firestore
    fun showData(){
        db.get().addOnSuccessListener {
            alFile.clear()
            for(doc in it){
                val hm = HashMap<String,Any>()
                hm.put(F_NAME,doc.get(F_NAME).toString())
                hm.put(F_TYPE,doc.get(F_TYPE).toString())
                hm.put(F_URL,doc.get(F_URL).toString())
                hm.put(F_DESC,doc.get(F_DESC).toString())
                alFile.add(hm)
            }
            adapter = CustomAdapter(this,alFile)
            b.listView.adapter = adapter
        }.addOnFailureListener { }
    }

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        b = ActivityMainBinding.inflate(layoutInflater)
        enableEdgeToEdge()
        setContentView(b.root)
    }
}
```

```
ViewCompat.setOnApplyWindowInsetsListener(  
    findViewById(b.main.id)  
) { v, insets ->  
    val systemBars = insets.getInsets(  
        WindowInsetsCompat.Type.systemBars()  
    )  
    v.setPadding(  
        systemBars.left, systemBars.top,  
        systemBars.right, systemBars.bottom  
    )  
    insets  
}  
b.btnUpPdf.setOnClickListener(this)  
b.btnUpImg.setOnClickListener(this)  
b.btnUpVid.setOnClickListener(this)  
b.btnUpWord.setOnClickListener(this)  
b.btnUpload.setOnClickListener(this)  
alFile = ArrayList()  
uri = Uri.EMPTY  
}  
  
override fun onStart() {  
    super.onStart()  
    //terhubung ke collection di Firestore bernama "files"  
    db = FirebaseFirestore.getInstance().collection("files")  
    db.addSnapshotListener { snapshot, error ->  
        error?.message?.let { Log.e("firestore", "error = $it") }  
    }  
    showData()  
}  
  
@SuppressLint("NewApi")  
override fun onClick(v: View?) {  
    val intent = Intent()  
    intent.action = Intent.ACTION_GET_CONTENT  
    when (v?.id) {  
        //menentukan tipe file yang akan diupload  
        R.id.btnUpWord -> {  
            fileType = "Docx"  
            intent.setType( //MIME  
                "application/vnd.openxmlformats-officedocument.wordprocessingml.document"  
            )  
        }  
  
        R.id.btnUpVid -> {  
            fileType = "Video"  
            intent.setType("video/*")  
        }  
  
        R.id.btnUpImg -> {  
            fileType = "Image"  
            intent.setType("image/*")  
        }  
  
        R.id.btnUpPdf -> {  
            fileType = "PDF"  
            intent.setType("application/pdf")  
        }  
    }
```

```
//button untuk upload file
R.id.btnUpload -> {
    lifecycleScope.launch { //coroutine
        try {
            contentResolver.openInputStream(uri).use {
                val file = it!!.readBytes()

                //memilih bucket bernama "pub_storage" di Supabase
                val bucket = supabase.storage.from("pub_storage")
                try {
                    val builder = AlertDialog.Builder(this@MainActivity)
                    builder.setMessage(
                        "Proses upload sedang berlangsung ... please wait")
                    builder.setCancelable(false)
                    dialog = builder.create()
                    dialog.show()

                    //mengupload file ke bucket
                    bucket.uploadAsFlow(
                        path = fileName,
                        data = file,
                        options = { upsert = false }).collect {
                        when (it) {
                            is UploadStatus.Progress -> {
                                //bagian ini bisa diisi dengan progress bar
                            }
                            is UploadStatus.Success -> { //jika upload sudah selesai maka ..
                                val hm = HashMap<String, Any>()
                                hm.put(F_NAME, fileName)
                                hm.put(F_TYPE, fileType)
                                hm.put(F_DESC, b.edFileDesc.text.toString())

                                //mendapatkan alamat URL dari file yang sudah diupload
                                hm.put(F_URL, bucket.publicUrl(fileName))

                                //menambahkan data ke Firestore
                                db.document(fileName).set(hm).addOnSuccessListener {
                                    if(dialog.isShowing) dialog.dismiss()
                                    showData()
                                }

                                b.txSelectedFile.setText("File yang akan diupload : ")
                                b.edFileDesc.setText("")
                            }
                        }
                    }
                } catch (e: UnknownRestException){
                    if(dialog.isShowing) dialog.dismiss()
                    b.txSelectedFile.setText("File yang akan diupload : ")
                    b.edFileDesc.setText("")
                    Toast.makeText(this@MainActivity,
                        "Tidak dapat terupload, file sudah ada di server",
                        Toast.LENGTH_LONG).show()
                }
            }
        } catch (e : FileNotFoundException){
            b.txSelectedFile.setText("File yang akan diupload : ")
            b.edFileDesc.setText("")
            Toast.makeText(this@MainActivity,
```

```
                "Pilih file yang akan diupload terlebih dahulu",  
                Toast.LENGTH_LONG).show()  
            }  
        }  
    }  
    if(v?.id != R.id.btnUpload) startActivityForResult(intent, RC_OK)  
}  
  
override fun onActivityResult(requestCode: Int, resultCode: Int, intent: Intent?) {  
    super.onActivityResult(requestCode, resultCode, intent)  
    if((resultCode == RESULT_OK) && (requestCode == RC_OK)){  
        if(intent != null){  
            uri = intent.data!!  
            Log.d("uri", "data = $uri")  
  
            if(uri.toString().contains("content:/")){  
                val cursor = contentResolver.query(uri, null,  
                    null, null, null)  
                val COL_FILE_NAME = cursor!!.getColumnIndex(  
                    OpenableColumns.DISPLAY_NAME)  
                cursor!!.moveToFirst()  
                fileName = cursor.getString(COL_FILE_NAME)  
                b.txSelectedFile.setText("File yang akan diupload : "+fileName)  
            }  
            if(uri.toString().contains("file:/")){  
                fileName = uri.lastPathSegment.toString() //mengambil nama file  
                b.txSelectedFile.setText(fileName)  
            }  
        }  
    }  
}
```

22. Tampilan akhir aplikasi seperti berikut ini

App Supa B

IMAGE

PDF

WORD

VIDEO

File yang akan diupload :

Deskripsi File :

Tuliskan deskripsi file yang akan diupload

Upload File

1000215412.jpg

Image

https://dufcskyuiwdcchgwaul.supabase.co/storage/v1/object/public/pub_storage/1000215412.jpg

ruang pertemuan

1000215435.jpg

Image

https://dufcskyuiwdcchgwaul.supabase.co/storage/v1/object/public/pub_storage/1000215435.jpg

1449. - Pemberitahuan dan Undangan Partisipasi Event Wattnext by PLN Mobile - PLN UID Jawa Tengah dan D.I Yogyakarta UP3 Purwokerto.pdf

PDF

https://dufcskyuiwdcchgwaul.supabase.co/storage/v1/object/public/pub_storage/1449. - Pemberitahuan dan Undangan Partisipasi Event Wattnext by PLN Mobile - PLN UID Jawa Tengah dan D.I Yogyakarta UP3 Purwokerto.pdf

Skema Penyediaan Informasi Akuntansi dan Perpajakan.docx

Docx

https://dufcskyuiwdcchgwaul.supabase.co/storage/v1/object/public/pub_storage/Skema Penyediaan Informasi Akuntansi dan Perpajakan.docx